Descriptive Statistical Analysis Template

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2025-11-18

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1 Data Loading and Preparation

```
# Load your dataset here
# Example: Replace with your actual data loading
# data <- read.csv("your_data.csv")
# data <- read.xlsx("your_data.xlsx")

# For demonstration, we'll use the built-in mtcars dataset
data <- mtcars
data$cyl <- factor(data$cyl)  # Convert to factor for categorical analysis
data$vs <- factor(data$vs)
data$am <- factor(data$am)

# Display basic information about the dataset
cat("Dataset dimensions:", dim(data), "\n")
```

```
## Dataset dimensions: 32 11
```

```
cat("Variable names:", names(data), "\n")
```

```
## Variable names: mpg cyl disp hp drat wt qsec vs am gear carb
```

```
str(data)
```

```
## 'data.frame':    32 obs. of  11 variables:
##  $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
##  $ cyl : Factor w/ 3 levels "4","6","8": 2 2 1 2 3 2 3 1 1 2 ...
##  $ disp: num  160 160 108 258 360 ...
##  $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
##  $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
##  $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
##  $ qsec: num  16.5 17 18.6 19.4 17 ...
##  $ vs  : Factor w/ 2 levels "0","1": 1 1 2 2 1 2 1 2 2 2 ...
##  $ am  : Factor w/ 2 levels "0","1": 2 2 2 1 1 1 1 1 1 1 ...
##  $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
##  $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
```

```
# Preview the data
kable(head(data, 10), caption = "First 10 rows of the dataset") %>%
  kable_styling(latex_options = c("striped", "hold_position"))
```

Table 1: First 10 rows of the dataset

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4

2 Frequency Tables

2.1 Absolute Frequency Tables

```
# For categorical variables
categorical_vars <- sapply(data, is.factor)
cat_data <- data[, categorical_vars, drop = FALSE]

if(ncol(cat_data) > 0) {
  for(var in names(cat_data)) {
    cat("\n### Absolute Frequency Table for", var, "\n")
    freq_table <- table(cat_data[[var]])
    print(kable(freq_table, col.names = c(var, "Frequency"),
                caption = paste("Absolute Frequency Table for", var)) %>%
          kable_styling(latex_options = c("striped", "hold_position")))
  }
} else {
  cat("No categorical variables found in the dataset.\n")
}
```

```
##
## ### Absolute Frequency Table for cyl
##
## \begin{longtable}[t]{lr}
## \caption{\label{tab:absolute-frequency-tables}Absolute Frequency Table for cyl}\\
## \toprule
## cyl & Frequency\\
## \midrule
## \cellcolor{gray!10}{4} & \cellcolor{gray!10}{11}\\
## 6 & 7\\
```

```
## \cellcolor{gray!10}{8} & \cellcolor{gray!10}{14}\\
## \bottomrule
## \end{longtable}
##
## ### Absolute Frequency Table for vs
##
## \begin{longtable}[t]{lr}
## \caption{\label{tab:absolute-frequency-tables}Absolute Frequency Table for vs}\\
## \toprule
## vs & Frequency\\
## \midrule
## \cellcolor{gray!10}{0} & \cellcolor{gray!10}{18}\\
## 1 & 14\\
## \bottomrule
## \end{longtable}
##
## ### Absolute Frequency Table for am
##
## \begin{longtable}[t]{lr}
## \caption{\label{tab:absolute-frequency-tables}Absolute Frequency Table for am}\\
## \toprule
## am & Frequency\\
## \midrule
## \cellcolor{gray!10}{0} & \cellcolor{gray!10}{19}\\
## 1 & 13\\
## \bottomrule
## \end{longtable}
```

```
# For numerical variables (binned)
numerical_vars <- sapply(data, is.numeric)
num_data <- data[, numerical_vars, drop = FALSE]

for(var in names(num_data)[1:2]) {  # Show first 2 numerical variables as example
  cat("\n### Binned Frequency Table for", var, "\n")
  breaks <- pretty(num_data[[var]], n = 8)
  freq_table <- table(cut(num_data[[var]], breaks = breaks, include.lowest = TRUE))
  print(kable(freq_table, col.names = c("Interval", "Frequency"),
              caption = paste("Binned Frequency Table for", var)) %>%
        kable_styling(latex_options = c("striped", "hold_position")))
}
```

```
##
## ### Binned Frequency Table for mpg
##
## \begin{longtable}[t]{lr}
## \caption{\label{tab:absolute-frequency-tables}Binned Frequency Table for mpg}\\
## \toprule
```

```
## Interval & Frequency\\
## \midrule
## \cellcolor{gray!10}{(10,15]} & \cellcolor{gray!10}{6}\\
## (15,20] & 12\\
## \cellcolor{gray!10}{(20,25]} & \cellcolor{gray!10}{8}\\
## (25,30] & 2\\
## \cellcolor{gray!10}{(30,35]} & \cellcolor{gray!10}{4}\\
## \bottomrule
## \end{longtable}
##
## ### Binned Frequency Table for disp
##
## \begin{longtable}[t]{lr}
## \caption{\label{tab:absolute-frequency-tables}Binned Frequency Table for disp}\\
## \toprule
## Interval & Frequency\\
## \midrule
## \cellcolor{gray!10}{(50,100]} & \cellcolor{gray!10}{5}\\
## (100,150] & 7\\
## \cellcolor{gray!10}{(150,200]} & \cellcolor{gray!10}{4}\\
## (200,250] & 1\\
## \cellcolor{gray!10}{(250,300]} & \cellcolor{gray!10}{4}\\
## \addlinespace
## (300,350] & 4\\
## \cellcolor{gray!10}{(350,400]} & \cellcolor{gray!10}{4}\\
## (400,450] & 1\\
## \cellcolor{gray!10}{(450,500]} & \cellcolor{gray!10}{2}\\
## \bottomrule
## \end{longtable}
```

2.2 Relative Frequency Tables

```r
# For categorical variables
if(ncol(cat_data) > 0) {
  for(var in names(cat_data)) {
    cat("\n### Relative Frequency Table for", var, "\n")
    freq_table <- table(cat_data[[var]])
    rel_freq_table <- prop.table(freq_table)
    rel_freq_df <- data.frame(
      Category = names(rel_freq_table),
      "Relative Frequency" = as.numeric(rel_freq_table),
      "Percentage" = as.numeric(rel_freq_table) * 100
    )
    print(kable(rel_freq_df, digits = 4,
                caption = paste("Relative Frequency Table for", var)) %>%
          kable_styling(latex_options = c("striped", "hold_position")))
```

```
  }
}
```

```
##
## ### Relative Frequency Table for cyl
##
## \begin{longtable}[t]{lrr}
## \caption{\label{tab:relative-frequency-tables}Relative Frequency Table for cyl}\\
## \toprule
## Category & Relative.Frequency & Percentage\\
## \midrule
## \cellcolor{gray!10}{4} & \cellcolor{gray!10}{0.3438} & \cellcolor{gray!10}{34.375}\\
## 6 & 0.2188 & 21.875\\
## \cellcolor{gray!10}{8} & \cellcolor{gray!10}{0.4375} & \cellcolor{gray!10}{43.750}\\
## \bottomrule
## \end{longtable}
##
## ### Relative Frequency Table for vs
##
## \begin{longtable}[t]{lrr}
## \caption{\label{tab:relative-frequency-tables}Relative Frequency Table for vs}\\
## \toprule
## Category & Relative.Frequency & Percentage\\
## \midrule
## \cellcolor{gray!10}{0} & \cellcolor{gray!10}{0.5625} & \cellcolor{gray!10}{56.25}\\
## 1 & 0.4375 & 43.75\\
## \bottomrule
## \end{longtable}
##
## ### Relative Frequency Table for am
##
## \begin{longtable}[t]{lrr}
## \caption{\label{tab:relative-frequency-tables}Relative Frequency Table for am}\\
## \toprule
## Category & Relative.Frequency & Percentage\\
## \midrule
## \cellcolor{gray!10}{0} & \cellcolor{gray!10}{0.5938} & \cellcolor{gray!10}{59.375}\\
## 1 & 0.4062 & 40.625\\
## \bottomrule
## \end{longtable}
```

```
# For numerical variables (binned)
for(var in names(num_data)[1:2]) {
  cat("\n### Relative Frequency Table for", var, "\n")
  breaks <- pretty(num_data[[var]], n = 8)
  freq_table <- table(cut(num_data[[var]], breaks = breaks, include.lowest = TRUE))
  rel_freq_table <- prop.table(freq_table)
```

```
  rel_freq_df <- data.frame(
    Interval = names(rel_freq_table),
    "Relative Frequency" = as.numeric(rel_freq_table),
    "Percentage" = as.numeric(rel_freq_table) * 100
  )
  print(kable(rel_freq_df, digits = 4,
              caption = paste("Relative Frequency Table for", var)) %>%
        kable_styling(latex_options = c("striped", "hold_position")))
}
```

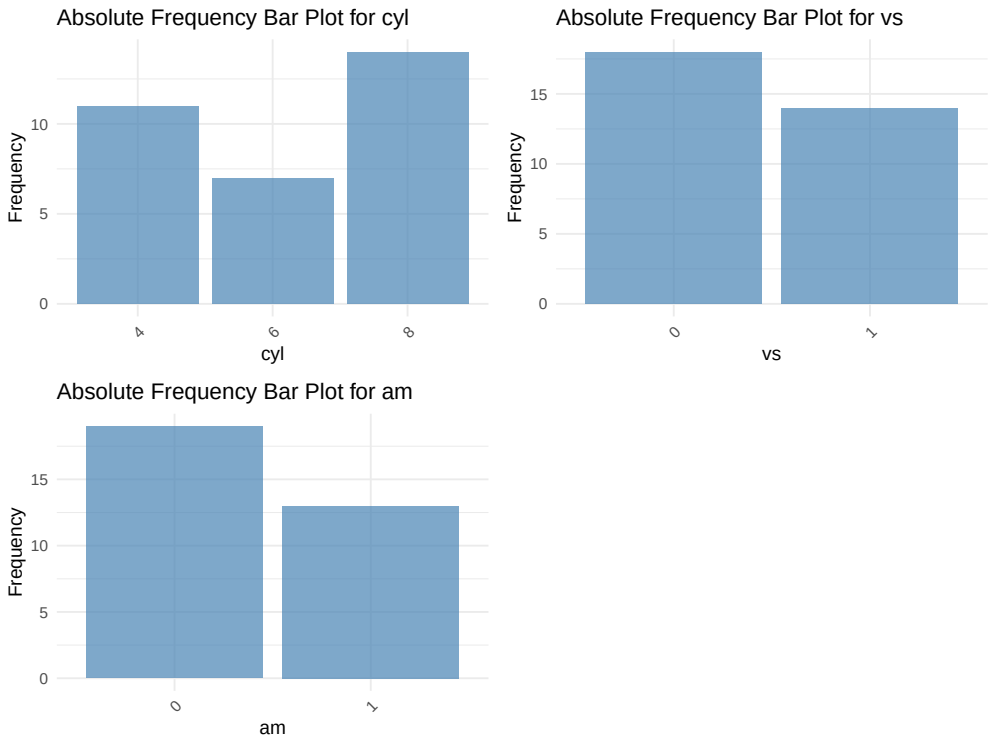
```
##
## ### Relative Frequency Table for mpg
##
## \begin{longtable}[t]{lrr}
## \caption{\label{tab:relative-frequency-tables}Relative Frequency Table for mpg}\\
## \toprule
## Interval & Relative.Frequency & Percentage\\
## \midrule
## \cellcolor{gray!10}{(10,15]} & \cellcolor{gray!10}{0.1875} & \cellcolor{gray!10}{18.75}\\
## (15,20] & 0.3750 & 37.50\\
## \cellcolor{gray!10}{(20,25]} & \cellcolor{gray!10}{0.2500} & \cellcolor{gray!10}{25.00}\\
## (25,30] & 0.0625 & 6.25\\
## \cellcolor{gray!10}{(30,35]} & \cellcolor{gray!10}{0.1250} & \cellcolor{gray!10}{12.50}\\
## \bottomrule
## \end{longtable}
##
## ### Relative Frequency Table for disp
##
## \begin{longtable}[t]{lrr}
## \caption{\label{tab:relative-frequency-tables}Relative Frequency Table for disp}\\
## \toprule
## Interval & Relative.Frequency & Percentage\\
## \midrule
## \cellcolor{gray!10}{(50,100]} & \cellcolor{gray!10}{0.1562} & \cellcolor{gray!10}{15.625}\
## (100,150] & 0.2188 & 21.875\\
## \cellcolor{gray!10}{(150,200]} & \cellcolor{gray!10}{0.1250} & \cellcolor{gray!10}{12.500}\
## (200,250] & 0.0312 & 3.125\\
## \cellcolor{gray!10}{(250,300]} & \cellcolor{gray!10}{0.1250} & \cellcolor{gray!10}{12.500}\
## \addlinespace
## (300,350] & 0.1250 & 12.500\\
## \cellcolor{gray!10}{(350,400]} & \cellcolor{gray!10}{0.1250} & \cellcolor{gray!10}{12.500}\
## (400,450] & 0.0312 & 3.125\\
## \cellcolor{gray!10}{(450,500]} & \cellcolor{gray!10}{0.0625} & \cellcolor{gray!10}{6.250}\\
## \bottomrule
## \end{longtable}
```

3 Bar Plots

3.1 Absolute Frequency Bar Plots

```
if(ncol(cat_data) > 0) {
  plots_list <- list()
  for(i in seq_along(names(cat_data))) {
    var <- names(cat_data)[i]
    p <- ggplot(cat_data, aes_string(x = var)) +
      geom_bar(fill = "steelblue", alpha = 0.7) +
      labs(title = paste("Absolute Frequency Bar Plot for", var),
           x = var, y = "Frequency") +
      theme_minimal() +
      theme(axis.text.x = element_text(angle = 45, hjust = 1))
    plots_list[[i]] <- p
  }

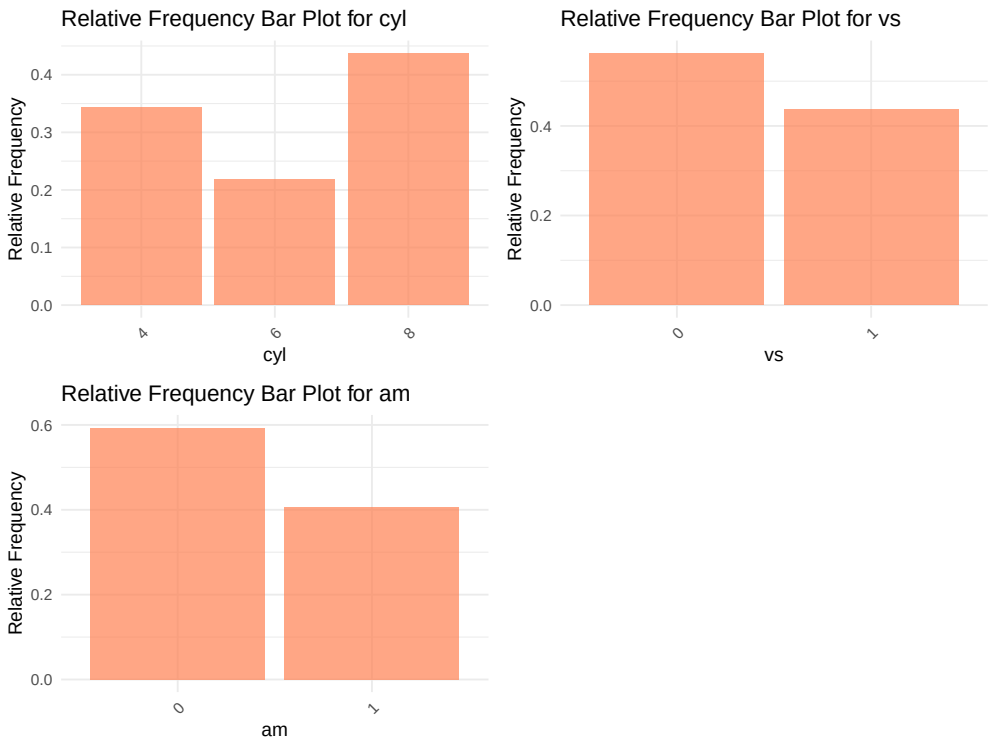
  if(length(plots_list) > 1) {
    do.call(grid.arrange, c(plots_list, ncol = 2))
  } else {
    print(plots_list[[1]])
  }
} else {
  cat("No categorical variables for bar plots.\n")
}
```



3.2 Relative Frequency Bar Plots

```
if(ncol(cat_data) > 0) {
  plots_list <- list()
  for(i in seq_along(names(cat_data))) {
    var <- names(cat_data)[i]
    p <- ggplot(cat_data, aes_string(x = var)) +
      geom_bar(aes(y = after_stat(count/sum(count))), fill = "coral", alpha = 0.7) +
      labs(title = paste("Relative Frequency Bar Plot for", var),
           x = var, y = "Relative Frequency") +
      theme_minimal() +
      theme(axis.text.x = element_text(angle = 45, hjust = 1))
    plots_list[[i]] <- p
  }

  if(length(plots_list) > 1) {
    do.call(grid.arrange, c(plots_list, ncol = 2))
  } else {
    print(plots_list[[1]])
  }
}
```



4 Summary Statistics

4.1 Measures of Center and Dispersion

```r
# Function to calculate comprehensive summary statistics
comprehensive_summary <- function(x) {
  if(is.numeric(x)) {
    stats <- c(
      Mean = mean(x, na.rm = TRUE),
      Median = median(x, na.rm = TRUE),
      Mode = as.numeric(names(sort(table(x), decreasing = TRUE))[1]),
      "Trimmed Mean (10%)" = mean(x, trim = 0.1, na.rm = TRUE),
      "Trimmed Mean (20%)" = mean(x, trim = 0.2, na.rm = TRUE),
      IQR = IQR(x, na.rm = TRUE),
      Variance = var(x, na.rm = TRUE),
      "Std Deviation" = sd(x, na.rm = TRUE),
      "Coeff of Variation" = sd(x, na.rm = TRUE) / mean(x, na.rm = TRUE) * 100,
      Skewness = skewness(x, na.rm = TRUE),
      Kurtosis = kurtosis(x, na.rm = TRUE),
      Min = min(x, na.rm = TRUE),
      Max = max(x, na.rm = TRUE),
      Range = diff(range(x, na.rm = TRUE))
    )
    return(round(stats, 4))
  } else {
    return(NULL)
  }
}

# Apply to all numerical variables
summary_stats <- sapply(num_data, comprehensive_summary)
summary_stats <- as.data.frame(summary_stats)

kable(summary_stats,
      caption = "Comprehensive Summary Statistics for Numerical Variables") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 2: Comprehensive Summary Statistics for Numerical Variables

	mpg	disp	hp	drat	wt	qsec	gear	carb
Mean	20.0906	230.7219	146.6875	3.5966	3.2172	17.8487	3.6875	2.8125
Median	19.2000	196.3000	123.0000	3.6950	3.3250	17.7100	4.0000	2.0000
Mode	10.4000	275.8000	110.0000	3.0700	3.4400	17.0200	3.0000	2.0000
Trimmed Mean (10%)	19.6962	222.5231	141.1923	3.5792	3.1527	17.8277	3.6154	2.6538

Trimmed Mean (20%)	19.2200	219.1750	137.9000	3.5755	3.1970	17.8175	3.5500	2.7000
IQR	7.3750	205.1750	83.5000	0.8400	1.0288	2.0075	1.0000	2.0000
Variance	36.3241	15360.7998	4700.8669	0.2859	0.9574	3.1932	0.5444	2.6089
Std Deviation	6.0269	123.9387	68.5629	0.5347	0.9785	1.7869	0.7378	1.6152
Coeff of Variation	29.9988	53.7178	46.7408	14.8664	30.4129	10.0116	20.0082	57.4293
Skewness	0.6404	0.4003	0.7614	0.2789	0.4438	0.3870	0.5546	1.1021
Kurtosis	2.7995	1.9103	3.0522	2.4351	3.1725	3.5538	2.0568	4.5361
Min	10.4000	71.1000	52.0000	2.7600	1.5130	14.5000	3.0000	1.0000
Max	33.9000	472.0000	335.0000	4.9300	5.4240	22.9000	5.0000	8.0000
Range	23.5000	400.9000	283.0000	2.1700	3.9110	8.4000	2.0000	7.0000

```
# Additional summary using psych package
describe_stats <- describe(num_data)
kable(describe_stats, digits = 4,
      caption = "Descriptive Statistics (psych package)") %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 3: Descriptive Statistics (psych package)

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	k
mpg	1	32	20.0906	6.0269	19.200	19.6962	5.4115	10.400	33.900	23.500	0.6107	-
disp	2	32	230.7219	123.9387	196.300	222.5231	140.4764	71.100	472.000	400.900	0.3817	-
hp	3	32	146.6875	68.5629	123.000	141.1923	77.0952	52.000	335.000	283.000	0.7260	-
drat	4	32	3.5966	0.5347	3.695	3.5792	0.7042	2.760	4.930	2.170	0.2659	-
wt	5	32	3.2172	0.9785	3.325	3.1527	0.7672	1.513	5.424	3.911	0.4231	-
qsec	6	32	17.8487	1.7869	17.710	17.8277	1.4159	14.500	22.900	8.400	0.3690	
gear	7	32	3.6875	0.7378	4.000	3.6154	1.4826	3.000	5.000	2.000	0.5289	-
carb	8	32	2.8125	1.6152	2.000	2.6538	1.4826	1.000	8.000	7.000	1.0509	

5 Box Plots

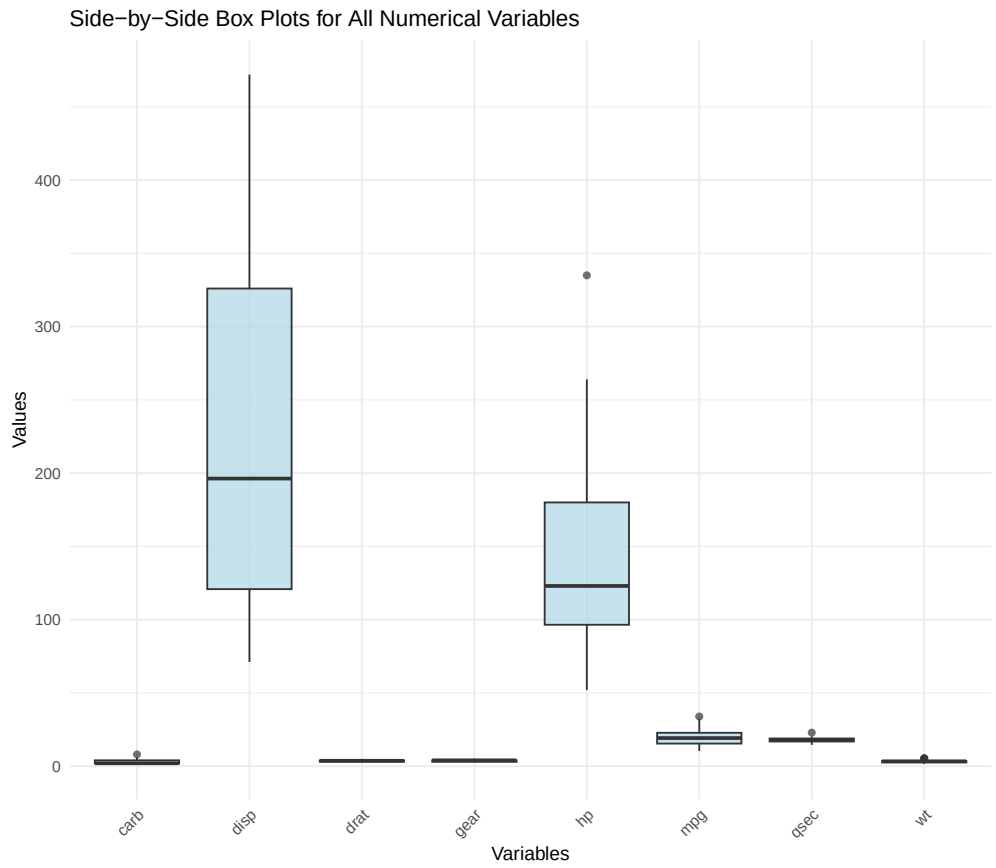
5.1 Side-by-Side Box Plots

```
# Box plots for numerical variables
if(ncol(num_data) > 0) {
  # Overall box plots
  num_data_long <- num_data %>%
    pivot_longer(everything(), names_to = "Variable", values_to = "Value")

  p1 <- ggplot(num_data_long, aes(x = Variable, y = Value)) +
    geom_boxplot(fill = "lightblue", alpha = 0.7) +
    labs(title = "Side-by-Side Box Plots for All Numerical Variables",
```

```
         x = "Variables", y = "Values") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))

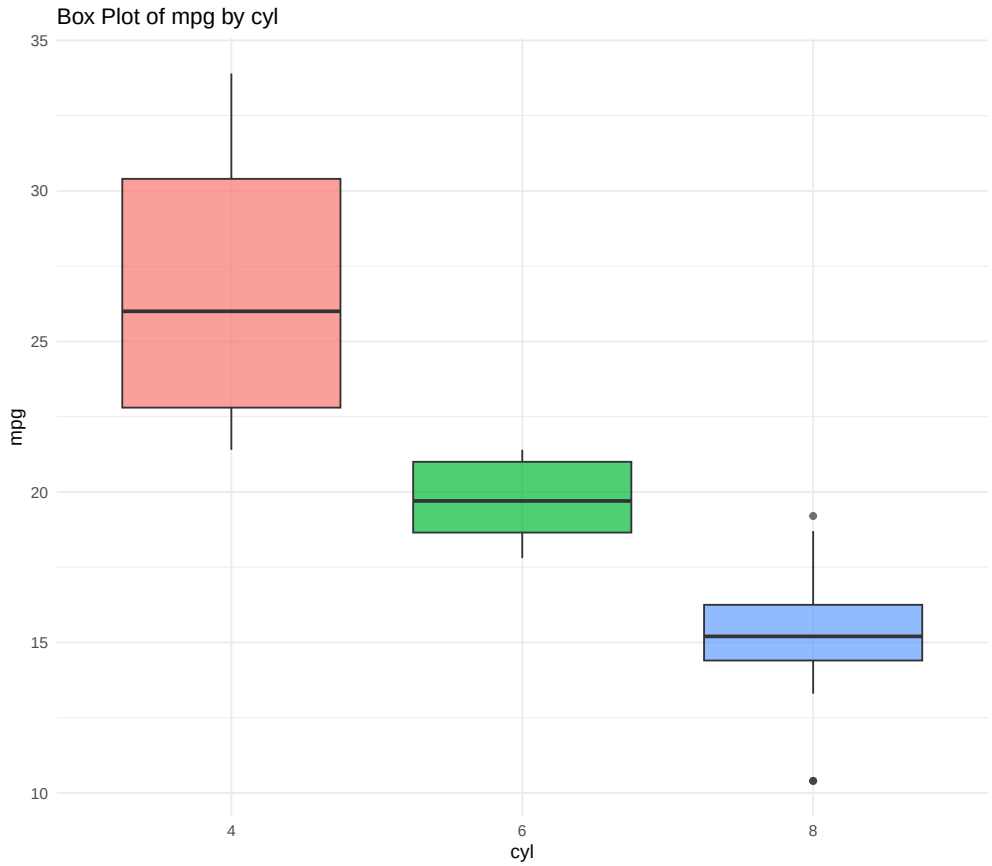
  print(p1)
}
```




```
# Box plots by categorical variables (if any)
if(ncol(cat_data) > 0 && ncol(num_data) > 0) {
  cat_var <- names(cat_data)[1]   # Use first categorical variable
  num_var <- names(num_data)[1]   # Use first numerical variable

  p2 <- ggplot(data, aes_string(x = cat_var, y = num_var, fill = cat_var)) +
    geom_boxplot(alpha = 0.7) +
    labs(title = paste("Box Plot of", num_var, "by", cat_var),
         x = cat_var, y = num_var) +
    theme_minimal() +
    theme(legend.position = "none")

  print(p2)
}
```



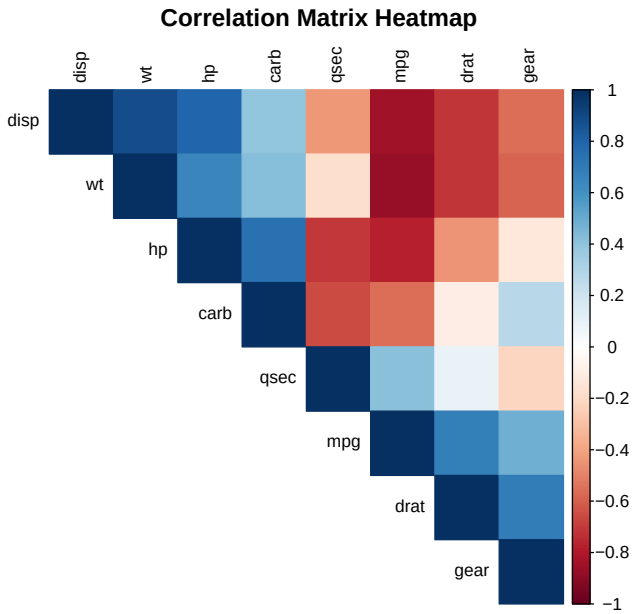
6 Scatter Plots and Correlations

6.1 Correlation Matrix

```
# Correlation matrix for numerical variables
if(ncol(num_data) > 1) {
  cor_matrix <- cor(num_data, use = "complete.obs")

  # Display correlation matrix as table
  kable(cor_matrix, digits = 3,
        caption = "Correlation Matrix") %>%
    kable_styling(latex_options = c("striped", "hold_position"))

  # Visual correlation matrix
  corrplot(cor_matrix, method = "color", type = "upper",
           order = "hclust", tl.cex = 0.8, tl.col = "black",
           title = "Correlation Matrix Heatmap", mar = c(0,0,2,0))
}
```



6.2 Scatter Plots

```
# if(ncol(num_data) > 1) {
#   # Create scatter plot matrix
#   pairs(num_data,
#         main = "Scatter Plot Matrix",
#         pch = 19,
#         col = alpha("steelblue", 0.6))
#
#   # Individual scatter plots for first few variable pairs
#   var_names <- names(num_data)[1:min(4, ncol(num_data))]
#   plots_list <- list()
#   plot_count <- 1
#
#   for(i in 1:(length(var_names)-1)) {
#     for(j in (i+1):length(var_names)) {
#       if(plot_count <= 6) {  # Limit to 6 plots
#         cor_val <- cor(num_data[[var_names[i]]], num_data[[var_names[j]]], use = "complete.o
#
#         p <- ggplot(data, aes_string(x = var_names[i], y = var_names[j])) +
#           geom_point(alpha = 0.6, color = "steelblue") +
#           geom_smooth(method = "lm", se = TRUE, color = "red") +
#           labs(title = paste(var_names[i], "vs", var_names[j],
#                              "\nCorrelation:", round(cor_val, 3))) +
#           theme_minimal()
#
#         plots_list[[plot_count]] <- p
#         plot_count <- plot_count + 1
```

```
#       }
#     }
#   }
#
#   if(length(plots_list) > 0) {
#     do.call(grid.arrange, c(plots_list, ncol = 2))
#   }
# }
```

7 Histograms

```
if(ncol(num_data) > 0) {
  plots_list <- list()

  for(i in seq_along(names(num_data))) {
    var <- names(num_data)[i]
    var_data <- num_data[[var]]

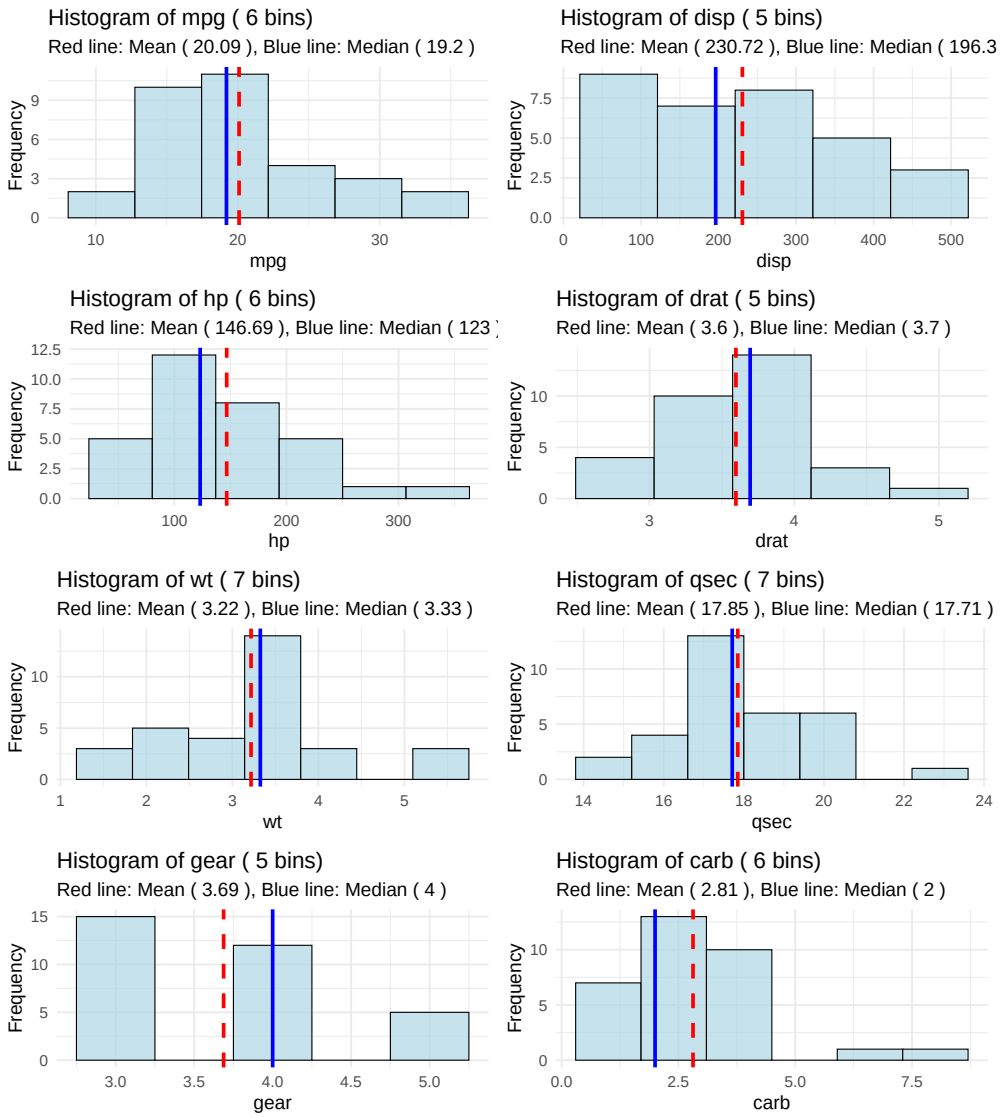
    # Calculate statistics
    mean_val <- mean(var_data, na.rm = TRUE)
    median_val <- median(var_data, na.rm = TRUE)

    # Determine optimal number of bins using Freedman-Diaconis rule
    iqr_val <- IQR(var_data, na.rm = TRUE)
    n <- length(var_data[!is.na(var_data)])
    bin_width <- 2 * iqr_val / (n^(1/3))
    n_bins <- ceiling(diff(range(var_data, na.rm = TRUE)) / bin_width)
    n_bins <- max(5, min(n_bins, 30))  # Ensure reasonable number of bins

    p <- ggplot(data, aes_string(x = var)) +
      geom_histogram(bins = n_bins, fill = "lightblue",
                     alpha = 0.7, color = "black", linewidth = 0.3) +
      geom_vline(xintercept = mean_val, color = "red",
                 linetype = "dashed", linewidth = 1) +
      geom_vline(xintercept = median_val, color = "blue",
                 linetype = "solid", linewidth = 1) +
      labs(title = paste("Histogram of", var, "(", n_bins, "bins)"),
           subtitle = paste("Red line: Mean (", round(mean_val, 2),
                            "), Blue line: Median (", round(median_val, 2), ")"),
           x = var, y = "Frequency") +
      theme_minimal()

    plots_list[[i]] <- p
  }
```

```
  # Arrange plots
  do.call(grid.arrange, c(plots_list, ncol = 2))
}
```



8 Quantile Plots

8.1 Q-Q Plots (Normal Quantile Plots)

```
if(ncol(num_data) > 0) {
  plots_list <- list()

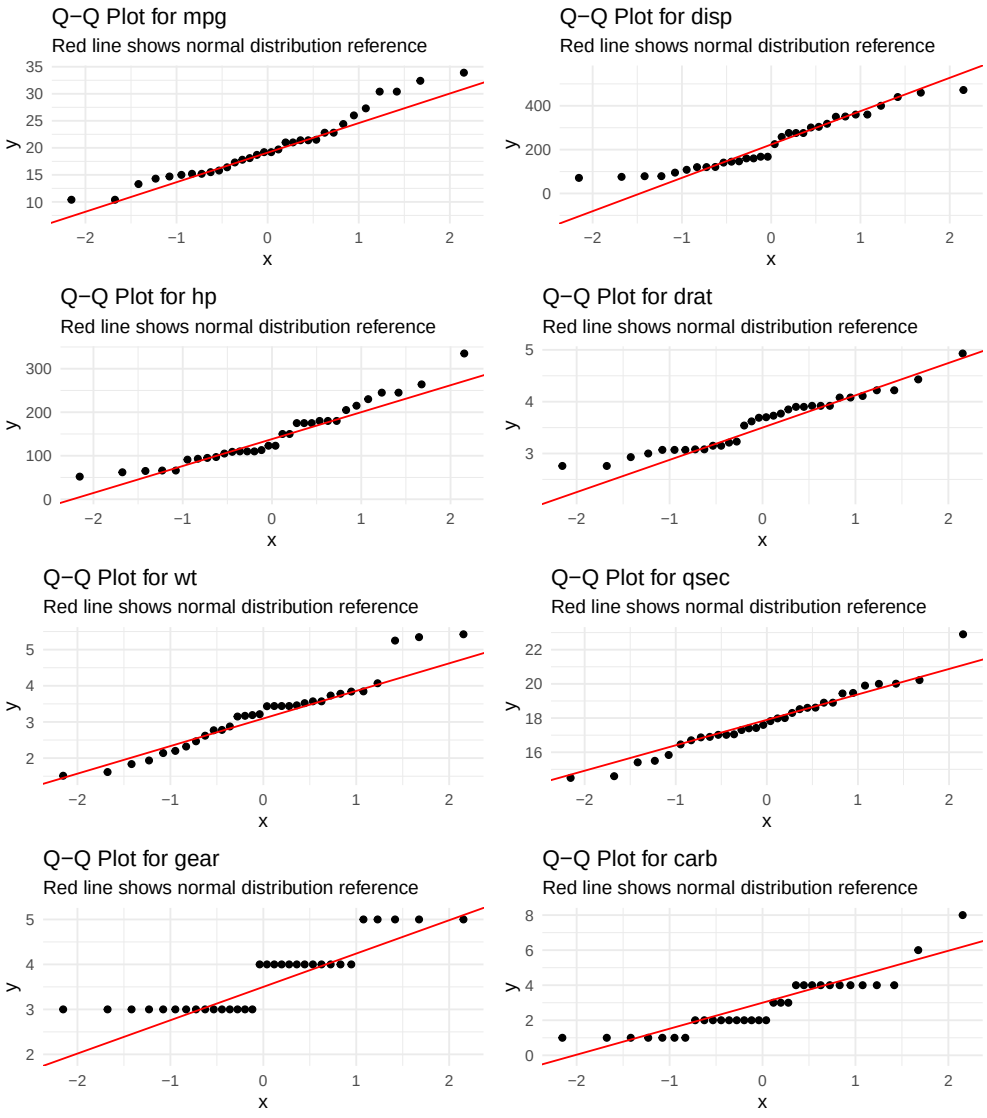
  for(i in seq_along(names(num_data))) {
```

```
    var <- names(num_data)[i]

    p <- ggplot(data, aes_string(sample = var)) +
      stat_qq() +
      stat_qq_line(color = "red") +
      labs(title = paste("Q-Q Plot for", var),
           subtitle = "Red line shows normal distribution reference") +
      theme_minimal()

    plots_list[[i]] <- p
  }

  do.call(grid.arrange, c(plots_list, ncol = 2))
}
```



8.2 Quantile-Quantile Plots Between Variables

```
# if(ncol(num_data) > 1) {
#   # Example: Compare first two numerical variables
#   var1 <- names(num_data)[1]
#   var2 <- names(num_data)[2]
#
#   ggplot(data, aes_string(x = var1, y = var2)) +
#     geom_qq() +
#     geom_abline(slope = 1, intercept = 0, color = "red") +
#     labs(title = paste("Quantile-Quantile Plot:", var1, "vs", var2),
#          subtitle = "Red line shows y=x reference") +
#     theme_minimal()
# }
```

9 Box-Cox Transformations

```
# Test for normality before Box-Cox transformation
normality_results <- data.frame(
  Variable = character(),
  Shapiro_p_value = numeric(),
  Normal = character(),
  stringsAsFactors = FALSE
)

if(ncol(num_data) > 0) {
  for(var in names(num_data)) {
    var_data <- num_data[[var]]
    var_data <- var_data[!is.na(var_data) & var_data > 0]  # Box-Cox requires positive values

    if(length(var_data) > 3 && length(var_data) <= 5000) {  # Shapiro-Wilk limitations
      shapiro_test <- shapiro.test(var_data)
      is_normal <- shapiro_test$p.value > 0.05

      normality_results <- rbind(normality_results, data.frame(
        Variable = var,
        Shapiro_p_value = round(shapiro_test$p.value, 6),
        Normal = ifelse(is_normal, "Yes (p > 0.05)", "No (p  0.05)")
      ))
    }
  }

  kable(normality_results,
        caption = "Normality Test Results (Shapiro-Wilk Test)") %>%
```

```
    kable_styling(latex_options = c("striped", "hold_position"))
}
```

Table 4: Normality Test Results (Shapiro-Wilk Test)

Variable	Shapiro_p_value	Normal
mpg	0.122881	Yes (p > 0.05)
disp	0.020807	No (p 0.05)
hp	0.048808	No (p 0.05)
drat	0.110061	Yes (p > 0.05)
wt	0.092655	Yes (p > 0.05)
qsec	0.593518	Yes (p > 0.05)
gear	0.000013	No (p 0.05)
carb	0.000438	No (p 0.05)

```
# Apply Box-Cox transformation to non-normal variables
if(ncol(num_data) > 0) {
  non_normal_vars <- normality_results$Variable[normality_results$Normal == "No (p  0.05)"]

  if(length(non_normal_vars) > 0) {
    cat("Applying Box-Cox transformations to non-normal variables:\n")

    plots_list <- list()
    transformation_summary <- data.frame(
      Variable = character(),
      Lambda = numeric(),
      stringsAsFactors = FALSE
    )

    for(var in non_normal_vars) {
      var_data <- num_data[[var]]
      var_data <- var_data[!is.na(var_data) & var_data > 0]

      if(length(var_data) > 0 && all(var_data > 0)) {
        # Find optimal lambda
        bc_result <- boxCox(lm(var_data ~ 1), plotit = FALSE)
        lambda <- bc_result$x[which.max(bc_result$y)]

        # Apply Box-Cox transformation
        if(abs(lambda) < 1e-6) {
          transformed_data <- log(var_data)
          transform_label <- "Log transformation (  0)"
        } else {
          transformed_data <- (var_data^lambda - 1) / lambda
          transform_label <- paste("Box-Cox transformation (  =", round(lambda, 3), ")")
```

```
      }

      # Store transformation info
      transformation_summary <- rbind(transformation_summary, data.frame(
        Variable = var,
        Lambda = round(lambda, 4)
      ))

      # Create before/after plots
      p1 <- ggplot(data.frame(x = var_data), aes(x = x)) +
        geom_histogram(bins = 20, fill = "lightcoral", alpha = 0.7) +
        labs(title = paste("Before:", var), x = var, y = "Frequency") +
        theme_minimal()

      p2 <- ggplot(data.frame(x = transformed_data), aes(x = x)) +
        geom_histogram(bins = 20, fill = "lightgreen", alpha = 0.7) +
        labs(title = paste("After:", transform_label),
             x = paste("Transformed", var), y = "Frequency") +
        theme_minimal()

      # Q-Q plots
      p3 <- ggplot(data.frame(x = var_data), aes(sample = x)) +
        stat_qq() + stat_qq_line(color = "red") +
        labs(title = paste("Q-Q Before:", var)) +
        theme_minimal()

      p4 <- ggplot(data.frame(x = transformed_data), aes(sample = x)) +
        stat_qq() + stat_qq_line(color = "red") +
        labs(title = paste("Q-Q After:", transform_label)) +
        theme_minimal()

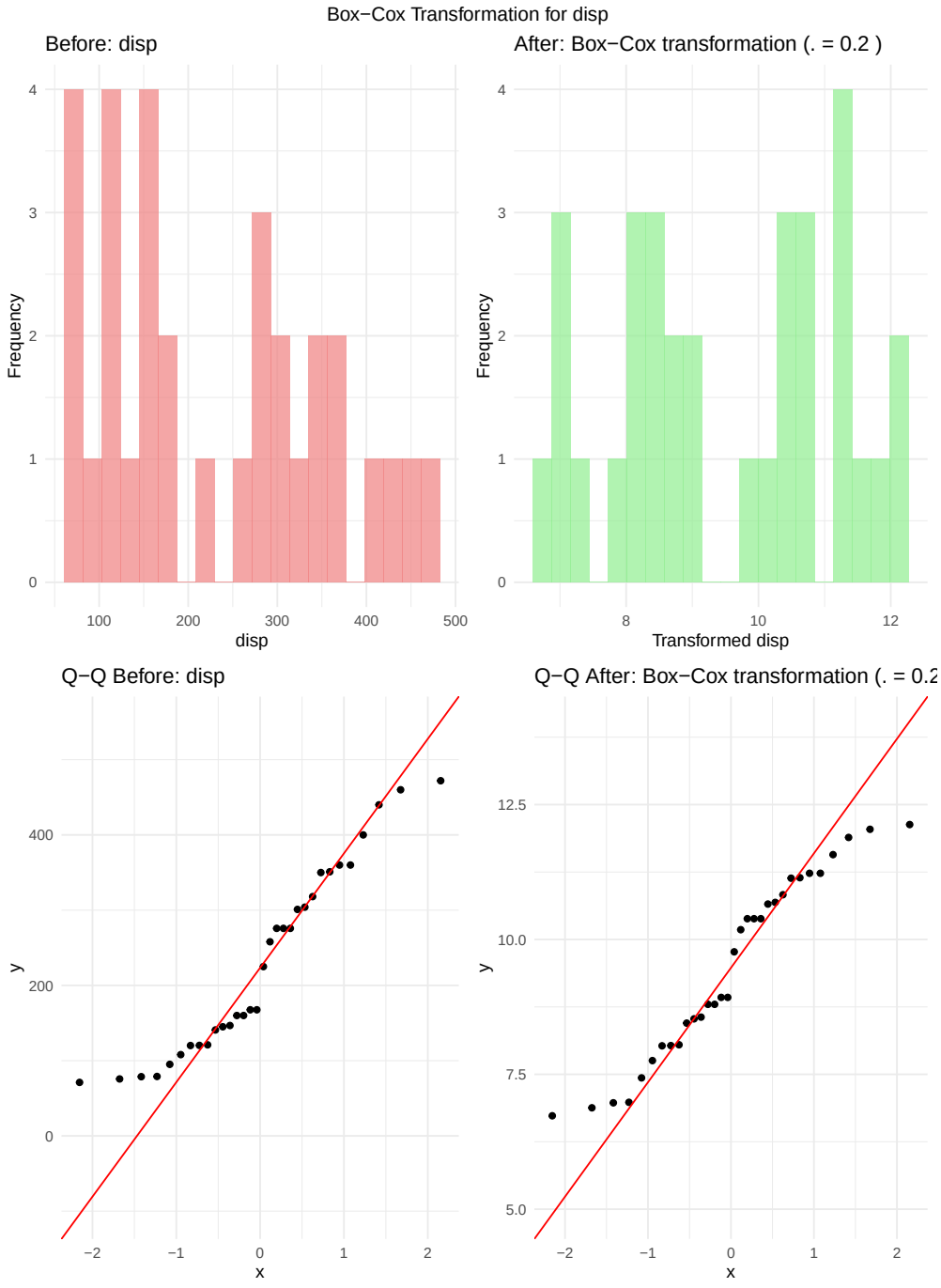
      # Combine plots
      combined_plot <- grid.arrange(p1, p2, p3, p4, ncol = 2,
                                    top = paste("Box-Cox Transformation for", var))
      plots_list[[length(plots_list) + 1]] <- combined_plot
    }
  }

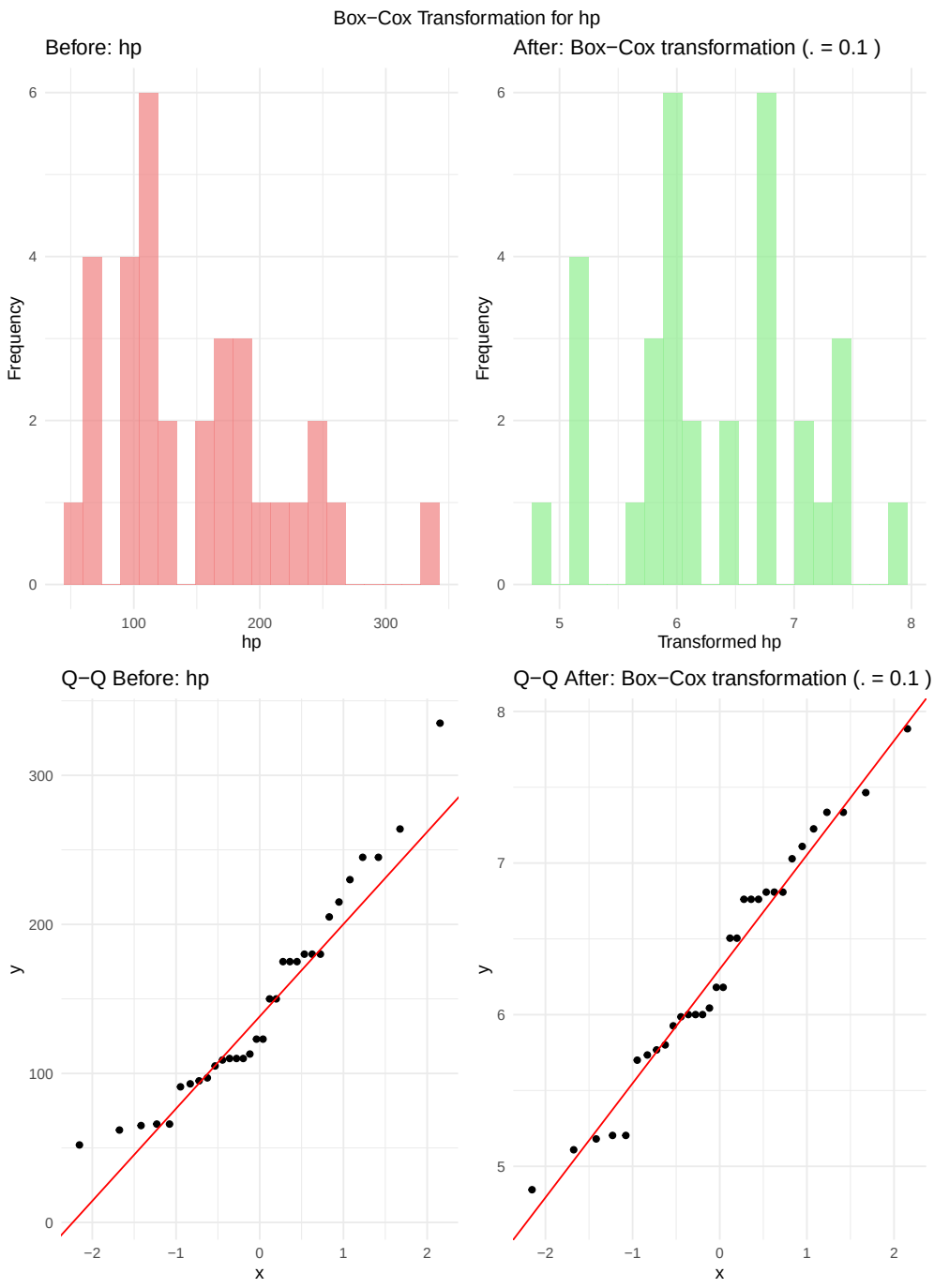
  # Display transformation summary
  if(nrow(transformation_summary) > 0) {
    kable(transformation_summary,
          caption = "Box-Cox Transformation Parameters") %>%
      kable_styling(latex_options = c("striped", "hold_position"))
  }

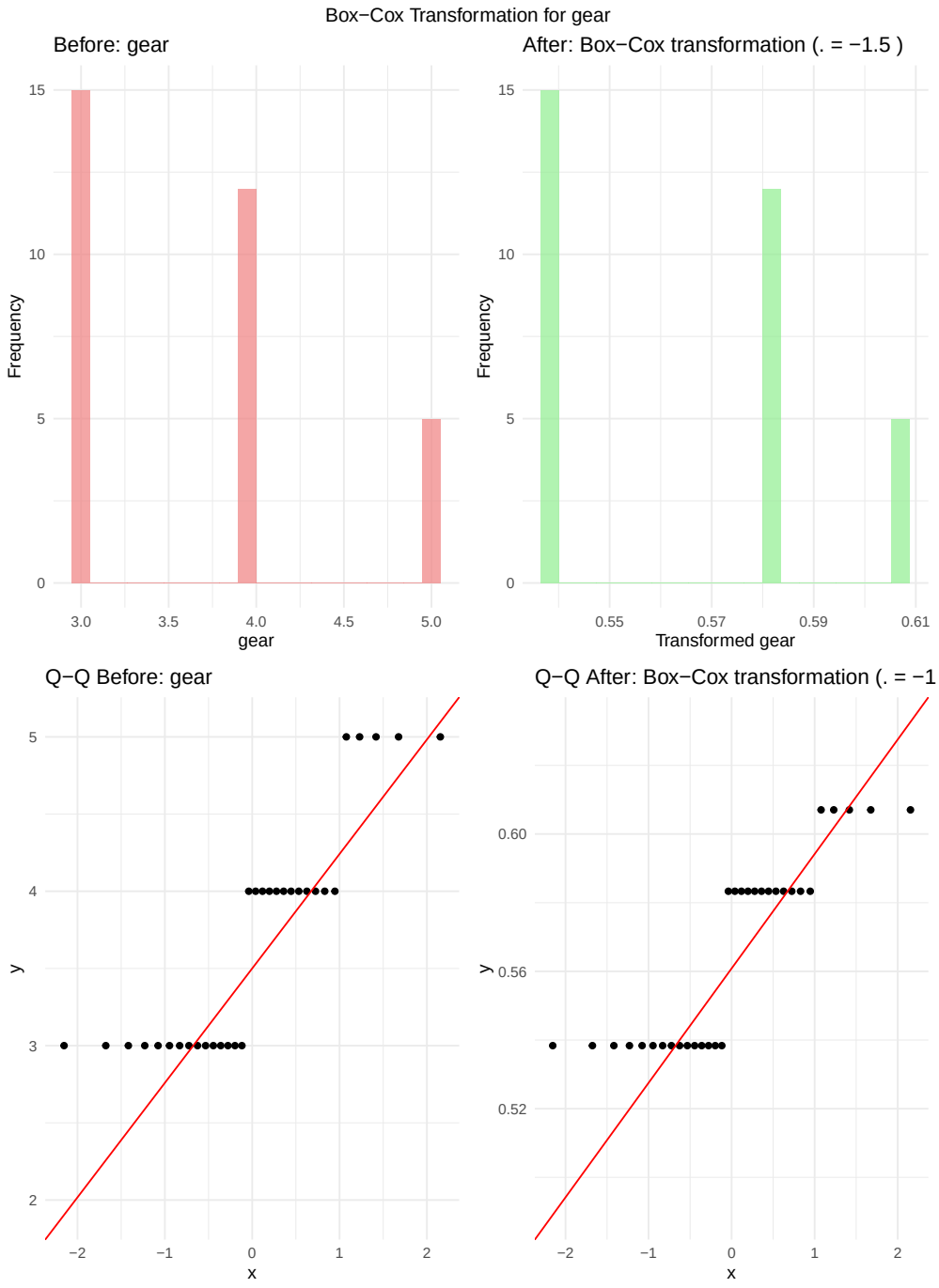
} else {
  cat("All variables appear to be normally distributed. No Box-Cox transformation needed.\n")
```

```
  }
} else {
  cat("No numerical variables available for Box-Cox transformation.\n")
}
```

```
## Applying Box-Cox transformations to non-normal variables:
```




Box−Cox Transformation for hp
Before: hp
After: Box−Cox transformation (. = 0.1)
Frequency
hp
Transformed hp
Q−Q Before: hp
Q−Q After: Box−Cox transformation (. = 0.1)
x
y


Box−Cox Transformation for gear
Before: gear
After: Box−Cox transformation (. = −1.5)
Frequency
gear
Transformed gear
Q−Q Before: gear
Q−Q After: Box−Cox transformation (. = −1
x
y

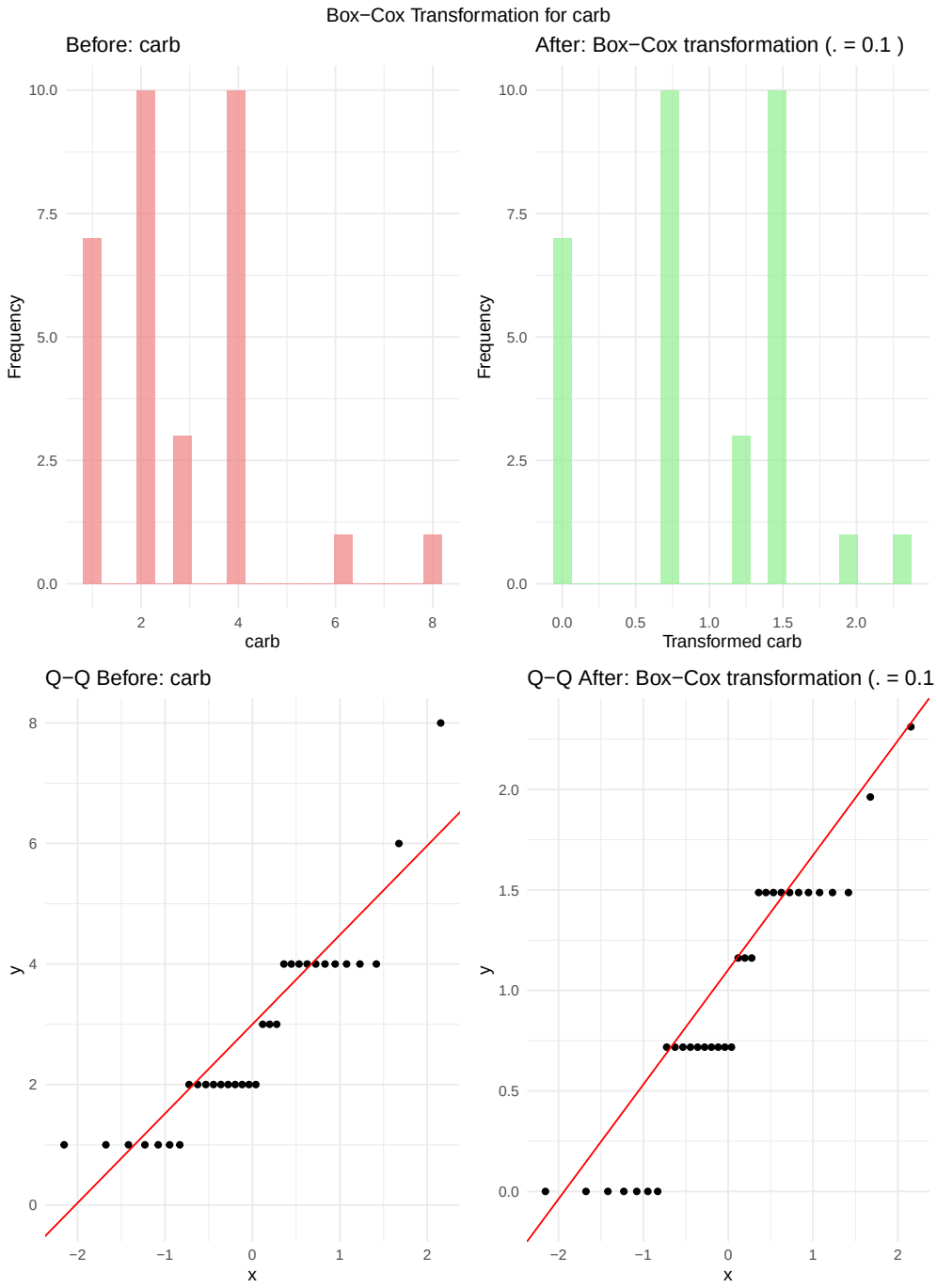


Table 5: Box-Cox Transformation Parameters

Variable	Lambda
disp	0.2
hp	0.1
gear	-1.5
carb	0.1

10 Summary and Conclusions

```
cat("## Analysis Summary\n\n")
```

```
## ## Analysis Summary
```

```
cat("Dataset:", deparse(substitute(data)), "\n")
```

```
## Dataset: data
```

```
cat("Total observations:", nrow(data), "\n")
```

```
## Total observations: 32
```

```
cat("Total variables:", ncol(data), "\n")
```

```
## Total variables: 11
```

```
cat("Numerical variables:", sum(sapply(data, is.numeric)), "\n")
```

```
## Numerical variables: 8
```

```
cat("Categorical variables:", sum(sapply(data, is.factor)), "\n")
```

```
## Categorical variables: 3
```

```
if(exists("normality_results") && nrow(normality_results) > 0) {
  normal_count <- sum(grepl("Yes", normality_results$Normal))
  cat("Variables following normal distribution:", normal_count, "out of", nrow(normality_result
}
```

```
## Variables following normal distribution: 4 out of 8
```

```
cat("\n## Key Findings:\n")
```

```
##
## ## Key Findings:
```

```r
cat("- Replace this section with your specific findings\n")
```

```
## - Replace this section with your specific findings
```

```r
cat("- Describe patterns observed in the data\n")
```

```
## - Describe patterns observed in the data
```

```r
cat("- Mention any outliers or unusual observations\n")
```

```
## - Mention any outliers or unusual observations
```

```r
cat("- Discuss correlations between variables\n")
```

```
## - Discuss correlations between variables
```

```r
cat("- Summarize transformation results if applicable\n")
```

```
## - Summarize transformation results if applicable
```

Note: This template provides a comprehensive framework for descriptive statistical analysis. Modify the data loading section and adjust the analyses based on your specific dataset and research questions. Remember to interpret the results in the context of your study domain.